

Medical Graph RAG: Evidence-based Medical Large Language Model via Graph Retrieval-Augmented Generation

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Abstract

We introduce MedGraphRAG, a novel graph-based Retrieval-Augmented Generation (RAG) framework designed to enhance LLMs in generating evidence-based medical responses, improving safety and reliability with private medical data. We introduce Triple Graph Construction and U-Retrieval to enhance GraphRAG, enabling holistic insights and evidence-based response generation for medical applications. Specifically, we connect user documents to credible medical sources and integrate Top-down Precise Retrieval with Bottom-up Response Refinement for balanced context awareness and precise indexing. Validated on 9 medical Q&A benchmarks, 2 health fact-checking datasets, and a long-form generation test set, MedGraphRAG outperforms state-of-the-art models while ensuring credible sourcing. Our code is publicly available.

1 Introduction

The rapid advancement of large language models (LLMs), such as OpenAI’s GPT-4 (OpenAI, 2023a), has accelerated research in natural language processing and driven numerous AI applications. However, these models still face significant challenges in specialized fields like medicine (Hadi et al., 2024; Williams et al., 2024; Xie et al., 2024). The first challenge is that these domains rely on vast knowledge bases -principles and notions discovered and accumulated over thousands of years; fitting such knowledge into the finite context window of current LLMs is a hopeless task. Supervised Fine-Tuning (SFT) provides an alternative to using the context window, but it is often prohibitively expensive or unfeasible due to the closed-source nature of most commercial models. Second, medicine is a specialized field that relies on a precise terminology system and numerous established truths, such as specific disease symptoms or drug side effects. In this domain, it is essential that LLMs do

not distort, modify, or introduce creative elements into the data. Unfortunately, verifying the accuracy of responses in medicine is particularly challenging for non-expert users. Therefore, the ability to perform complex reasoning using large external datasets, while generating accurate and credible responses backed by verifiable sources, is crucial in medical applications of LLMs.

Retrieval-augmented generation (RAG) (Lewis et al., 2021) is a technique that answers user queries using specific and private datasets without requiring further training of the model. However, RAG struggles to synthesize new insights and underperforms in tasks requiring a holistic understanding across extensive documents. GraphRAG (Hu et al., 2024) has been recently introduced to overcome these limitations. GraphRAG constructs a knowledge graph from raw documents using an LLM, and retrieves knowledge from the graph to enhance responses. By representing clear conceptual relationships across the data, it significantly outperforms classic RAG, especially for complex reasoning (Hu et al., 2024). However, its graph construction lacks a specific design to ensure response authentication and credibility, and its hierarchical community construction process is costly, as it is designed to handle various cases for general-purpose use. We find that specific effort is required to apply it effectively in the medical domain.

In this paper, we introduce a novel graph-based RAG method for medical domain, which we refer to as Medical GraphRAG (MedGraphRAG). This technique enhances LLM performance in the medical domain by generating evidence-based responses and official medical term explanation, which not only increases their credibility but also significantly improves their overall quality. Our method builds on GraphRAG with a more sophisticated graph construction technique, called Triple Graph Construction, to generate evidence-based responses, and an efficient retrieval method, U-Retrieval, which im-

proves response quality with few costs. In Triple Graph Construction, we design a mechanism to link user RAG data to credible medical papers and foundational medical dictionaries. This process generates triples $[RAG\ data, source, definition]$ to construct a comprehensive graph of user documents. It enhances LLM reasoning and ensures responses are traceable to their sources and definitions, guaranteeing reliability and explainability. We also developed a unique U-Retrieval strategy to respond to user queries. Instead of building costly graph communities, we streamline the process by summarizing each graph using predefined medical tags, then iteratively clustering similar graphs to form a multi-layer hierarchical tag structure, from broad to detailed tags. The LLM generates tags for the user query and indexes the most relevant graph based on tag similarity in a top-down approach, using it to formulate the initial response. Then it refines the response by progressively integrating back the higher-level tags in a bottom-up manner until the final answer is generated. This U-Retrieval technique strikes a balance between global context awareness and the retrieval efficiency.

To evaluate our MedGraphRAG method, we implemented it on several popular open-source and commercial LLMs, including GPT (OpenAI, 2023b), Gemini (Team et al., 2023) and LLaMA (Touvron et al., 2023). The results evaluated across 9 medical Q&A benchmarks show that MedGraphRAG yields materially better results than classic RAG and GraphRAG. Our final results even surpass many specifically trained LLMs on medical corpora, setting a new state-of-the-art (SOTA) across all benchmarks. To verify its evidence-based response capability, we quantitatively tested MedGraphRAG on 2 health fact-checking benchmarks and conducted a human evaluation by experienced clinicians. Both evaluations strongly support that our responses are more source-based and reliable than previous methods.

Our contributions are as follows:

1. We are the first to propose a specialized framework for introducing graph-based RAG in the medical domain, which we named MedGraphRAG.
2. We have developed unique Triple Graph Construction and U-Retrieval methods that enable LLMs to efficiently generate evidence-based responses utilizing holistic RAG data.
3. MedGraphRAG outperforms other retrieval methods and extensively fine-tuned Medical LLMs across a wide range of medical Q&A benchmarks,

establishing the new SOTAs.

4. Validated by human evaluations, MedGraphRAG is able to generate more understandable and evidence-based responses in the medical domain.

2 Method

The overall workflow of MedGraphRAG is shown in Fig. 1. We first construct the knowledge graphs from the documents by using Triple Graph Construction (Section 2.1), then tag the graphs for U-Retrieval to response the user queries (Section 2.2).

2.1 Triple Graph Construction

2.1.1 Preliminary: Document Chunking & Entities Extraction

Large medical documents often contain diverse content. We segment them into chunks respecting LLMs’ context limits. We adopt the semantic chunking function implemented in LangChain to chunk the documents (langchain, 2024). Specifically, we isolate paragraphs P_i within the document $D = \{P_1, P_2, \dots, P_{N_p}\}$ using a text embedding model. We then set a buffer size of 5 and enforce the token limit according to the graph construction LLM $\mathcal{L}^G$.

We then extract entities from each chunk through graph construction LLM $\mathcal{L}^G$. We prompt $\mathcal{L}^G$ to identify all relevant entities $E = \{e_1, e_2, \dots, e_{N_e^1}\}$ in each chunk and generate a structured output with *name*, *type*, and *a description of the context*: $e = \{na, ty, cx\}$, as the examples shown in the Step2 in Fig. 1. We set *name* be the text from the document, *type* selected from the UMLS semantic types (Bodenreider, 2004), and *context* a few sentences generated by $\mathcal{L}^G$ contextualized within the document.

2.1.2 Triple Linking

Medicine relies on precise terminology and established facts, making it essential for LLMs to produce responses grounded in established facts. To achieve this, we introduced Triple Graph Construction, linking user documents to credible sources and professional definitions. Specifically, we build repository graph (RepoGraph), which is intended to be fixed across different users, providing established sources and controlled vocabulary definitions for user RAG documents. We construct RepoGraph under user RAG graph with two layers: one based on medical papers/books and another based on medical dictionaries. We build the bottom

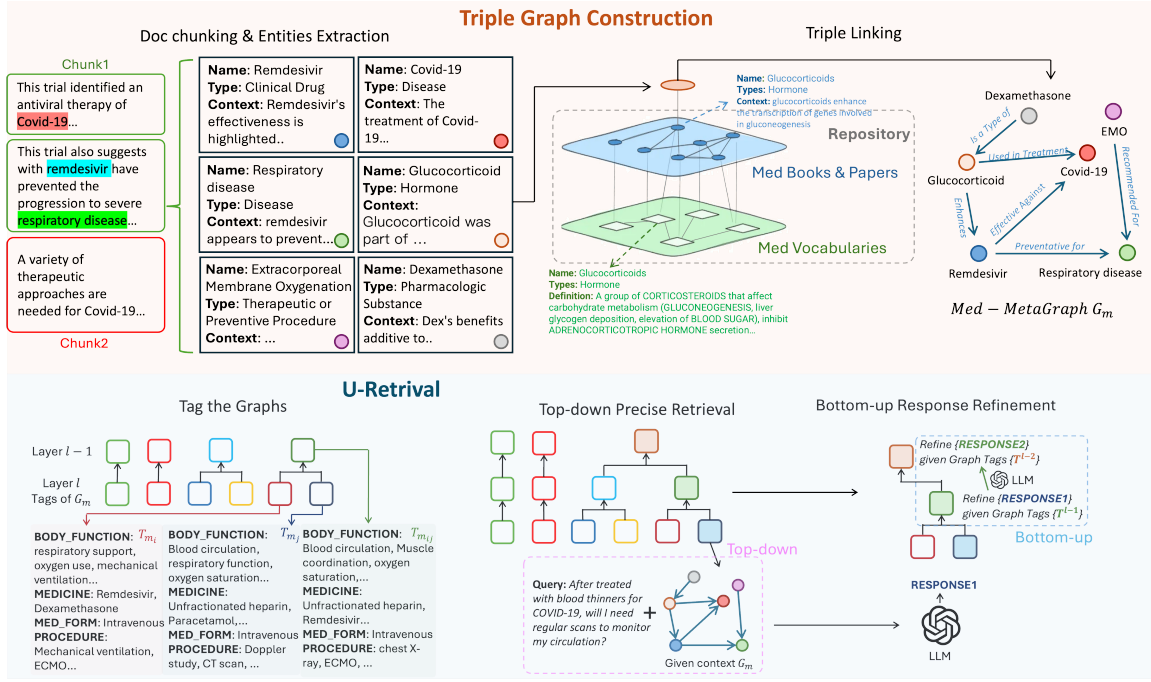


Figure 1: The overall workflow of MedGraphRAG begins with Triple Graph Construction, where documents are chunked, and entities are extracted. Triple linking then connects user entities to referenced papers and vocabulary graph layers, forming the Med-MetaGraph. In the subsequent U-Retrieval phase, graphs are tagged to enable top-down precise retrieval and bottom-up response refinement, ensuring graph-enhanced query responses.

layer of RepoGraph as UMLS (Bodenreider, 2004) graph, which consist comprehensive, well-defined medical vocabularies and their relationships. The upper layer of RepoGraph is constructed from medical textbooks and scholarly articles using the same graph construction method described here.

The entities of all three tiers of graphs are hierarchically linked through semantic relationships. Let us denoted entities extracted from RAG documents as E^1 . We link them to entities extracted from medical books/papers, denoted as E^2 , based on their relevance, which is determined by computing the cosine similarity between their content embeddings $\phi(C_e)$. The content of an entity C_e is the concatenation of its *name*, *type*, and *context*, represented as: $C_e = \text{Text}[\text{name: na; type: ty; context: cx}]$. This directed linking is annotated as *the reference of*, indicating the reference relationship between entities in the two layers:

$$R_{e_2}^1 = \left\{ (e_1^1, \text{TheReferenceOf}, e_2^2) \mid \frac{\phi(C_{e_1^1}) \cdot \phi(C_{e_2^2})}{\|\phi(C_{e_1^1})\| \|\phi(C_{e_2^2})\|} \geq \delta_r \right\},$$

where δ_r is the pre-defined threshold. Entities $e^2 \in E^2$ are linked to $e^3 \in E^3$ through the same way with relationships annotated as *the definition of*. Thus, RAG entities are constructed as triples [RAG entity, source, definition].

We then instruct $\mathcal{L}^G$ to identify the relationships among RAG entities in each chunk, which we noted as $e^1 \in E_m$. This relationship is a concise phrase generated by $\mathcal{L}^G$ based on the content of the entity C_{e^1} and associated references $\{C_{e^2} | R_{e^2}^1 = \text{the reference of}\}$. The identified relationships specify the source and target entities, provide a description of their relationship: $R_{e_i^1}^j = \{(e_i^1, r_{ij}, e_j^1) \mid r_{ij} = \mathcal{L}_{rel}^G(C_{e_i^1}; C_{e_j^2}, C_{e_j^1}; C_{e_j^2})\}$, where $\mathcal{L}_{rel}^G$ is $\mathcal{L}^G$ with relationship identification and generation prompt. We show an example of relationship linking in the Step4 of Fig. 1. After performing this analysis, we have generated a directed graph for each data chunk, which is referred to as Meta-MedGraphs $G_m = \{E_m, R(E_m)\}$.

2.2 U-Retrieval

2.2.1 Preliminary: Graph Tagging

Organizing and summarizing the graphs in advance is intuitive and has proven to facilitate efficient retrieval (Hu et al., 2024). However, unlike GraphRAG, we avoid constructing costly graph communities. We observe that, unlike general language content, medical text is often structured and can be summarized effectively using predefined tags. Motivated by this, we simply summarize

each Meta-MedGraph G_m with several predefined tags T , and iteratively generate more abstract tag summaries for clusters of closely-related graphs. Specifically, LLM $\mathcal{L}^G$ first summarises the content of each Meta-MedGraph $\{C_e \mid e \in G_m\}$ given a set of given tags T . The tags T consist of multiple medical categories following Society for Testing and Materials (ASTM) standards for content of electronic health records, mainly including *Symptoms*, *Patient History*, *Body Functions*, and *Medication*. This process generates a structured tag-summary for each G_m , denoted as T_m .

We then apply a variant agglomerative hierarchical clustering method with dynamic thresholding based on tag similarity, to group the graphs and generate synthesized tag summaries. Initially, each graph begins as its own group. At each iteration, we compute the tag similarity between all pairs of clusters and dynamically set the threshold δ_t to merge the top 20% most similar pairs. The graphs will be merged if all pairwise similarities within the group exceed δ_t . Note that we don't really link the nodes across different graphs, but generate a synthesized tag-summary for each group. Specifically, we calculate the similarity of pairs by measuring the average cosine similarity of all their tag embeddings. Let $\phi(t)$ denote the embedding of a tag $t \in T_m$. Taking two Meta-MedGraphs G_{m_i} and G_{m_j} with tag sets T_{m_i} and T_{m_j} as an example, we generate the abstract tag summary $T_{m_{ij}}$ if their cosine similarity of tag embeddings $\phi(t)$ and $\phi(t')$ higher than the threshold δ_t

$$T_{m_{ij}} = \mathcal{L}^G(T_{m_i}, T_{m_j}), \quad \text{if} \\ \frac{1}{|T_{m_i}| \cdot |T_{m_j}|} \sum_{t \in T_{m_i}} \sum_{t' \in T_{m_j}} \frac{\phi(t)^\top \phi(t')}{\|\phi(t)\| \|\phi(t')\|} \geq \delta_t;$$

These newly merged tag-summary, along with those that remain unmerged, form a new layer of tags. As tag-summaries become less detailed at higher layers, there is a trade-off between precision and efficiency. In practice, we limit the process to 12 layers, as this is sufficient for most model variants (detailed in Fig. 5).

2.2.2 Top-down Precise Retrieval

After constructing the graph, we use response LLM $\mathcal{L}^R$ efficiently retrieves information to respond to user queries. We begin by generating tag-summary on the user query $T_Q = \mathcal{L}^R(Q)$, and use these to identify the most relevant graph through a Top-down Precise Retrieval. Let's indicate the j^{th} tags

at layer i summarised tag T^i as $T^i[j]$, it starts from the top layer: T^0 , progressively indexing down by selecting the most similar tag in each layer:

$$T^{i+1} = \arg \max_{T^i[j] \in T^i} \text{sim}(T_Q, T^i[j])$$

until we reach the tag for the target Meta-MedGraph G_{m_t} . We then retrieve Top N_u entities based on the embedding similarity between the query and the entity content: $E_r = \{e \mid \text{Top}N_u(\text{sim}(\phi(Q), \phi(C_e))), e \in M_t\}$, and gather all their Top k_u nearest triple neighbours $Tr_i^{\leq k_u}(e)$ as $E_r^{k_u} = \{e, Tr_i^{\leq k_u}(e), \mid e \in E_r\}$.

2.2.3 Bottom-up Response Refinement

By using all these entities and their relationships $G_r = \{E_r^{k_u}, R(E_r^{k_u})\}$, we prompt $\mathcal{L}^R$ to answer the question given the concatenated entity *names* and *relationships* in G_r : *Given QUESTION: $\{Q\}$. GRAPH: $\{e_i[na] + R_{e_i}^{e_j} + e_j[na], \dots\}$. Answer the user question: QUESTION using the graph: GRAPH... as $\mathcal{L}_{G_r}^R$.*

In the Bottom-up Response Refinement step, we then move back to the higher-level tag retrieved in the previous step T^{i-1} , in a bottom-up manner. We provide $\mathcal{L}^R$ *QUESTION: $\{Q\}$, LAST RESPONSE: ..., and SUMMARY: $\{T^{i-1}\}$, and ask it to Adjust the response: RESPONSE of the question: QUESTION using the updated information: SUMMARY.* $\mathcal{L}^R$ continues refining its responses until it reaches the target layer. In practice, we retrieve 4-6 layers depends on the baseline LLM, a detailed experiment about it is shown in Fig. 5. It ultimately generate a final response after scanning all indexed graphs along the trajectory. This method enables the LLM to gain a comprehensive overview by interacting with all relevant data in the graph, while remaining efficient by accessing less relevant data in summarized form.

3 Experiment

3.1 Dataset

3.1.1 RAG data

We anticipate that users will use frequently-updated private data as RAG data, such as patient electronic medical records. Thus, we employ MIMIC-IV (Johnson et al., 2023), a publicly available electronic health record dataset, as RAG data.

3.1.2 Repository data

We provide repository data to support LLM responses with credible sources and authoritative vo-

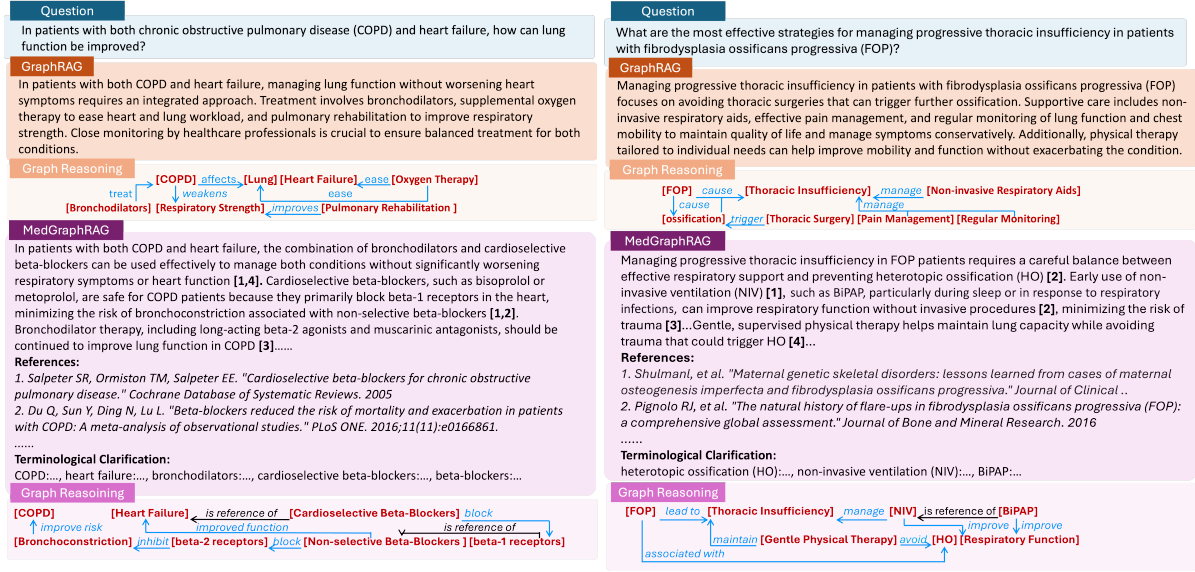


Figure 2: Example responses from GraphRAG and MedGraphRAG, with abstracted graphs. MedGraphRAG provides more detailed explanations and more complex reasoning with evidences. Full results are in the appendix.

cabulary definitions. We use MedC-K (Wu et al., 2023), a corpus containing 4.8 million biomedical academic papers and 30,000 textbooks, along with all evidence publications from FakeHealth (Dai et al., 2020) and PubHealth (Kotonya and Toni, 2020), as the upper repository data, and UMLS graph, which includes authoritative medical vocabularis and semantic relationships as the bottom repository data.

3.1.3 Test Data

Our test set are the test split of 9 multiple-choice biomedical datasets from the MultiMedQA suite, 2 fact verification datasets about public health, i.e., FakeHealth (Dai et al., 2020) and PubHealth (Kotonya and Toni, 2020), and 1 test set we collected, called DiverseHealth. MultiMedQA includes MedQA (Jin et al., 2021), MedMCQA (Pal et al., 2022) PubMedQA (Jin et al., 2019) and MMLU clinic topics (Hendrycks et al., 2020). We also collected the DiverseHealth test set, focused on health equity, consisting of 50 real-world clinical questions that cover a wide range of topics, including rare diseases, minority health, comorbidities, drug use, alcohol, COVID-19, obesity, suicide, and chronic disease management.

3.2 Experiment Setting

We compare different RAG methods across 6 language models as $\mathcal{L}^R$: Llama2 (13B, 70B), Llama3 (8B, 70B), Gemini-pro, and GPT-4. The Llama models were obtained from their official Hugging-

Face page. We used *gemini-1.0-pro* for Gemini-pro and *gpt-4-0613* for GPT-4. We primarily compare our approach with standard RAG implemented by LangChain(langchain, 2024) and GraphRAG (Edge et al., 2024a) implemented by Microsoft Azure (microsoft, 2024). All retrieval methods are compared under same RAG data and test data.

We deploy $\mathcal{L}^G$ as *Llama3-70B* to construct the graph. For text embeddings, we utilize OpenAI’s *text-embedding-3-large* model. Model comparison is performed using a 5-shot response ensemble (Li et al., 2024). MedGraphRAG used U-Retrieval with 4 levels on GPT-4, and 5 levels for the others. In the retrieval, we picked top 60 entities with their 16-hop neighbors. Unless otherwise noted, all thresholds are set as 0.5. We use the same query prompt for all models to generate responses.

3.3 Results

3.3.1 Multi-Choice Evaluation

Baselines with different retrievals First, we conducted experiments to evaluate retrieval methods on various LLM baselines, with the results shown in Table 1. We compared MedGraphRAG against baselines without retrieval, standard RAG, and GraphRAG. Performance is measured by the accuracy of selecting the correct option. The results show that MedGraphRAG significantly enhances LLM performance on both health fact-checking and medical Q&A benchmarks. Compared to baselines without retrieval, MedGraphRAG achieves

an average improvement of nearly 10% in fact-checking and 8% in medical Q&A. When compared to baselines using GraphRAG, it demonstrates an average improvement of around 8% in fact-checking and 5% in medical Q&A. Notably, MedGraphRAG yields more pronounced improvements in smaller LLMs, such as *Llama2_{13B}* and *Llama2_{8B}*. This suggests that MedGraphRAG effectively utilizes the models’ own reasoning capabilities while providing them with additional knowledge beyond their parameters, serving as an external memory for information.

Comparing with SOTA Medical LLMs When applied MedGraphRAG to larger models, like *Llama_{70B}* or GPT, it resulted in new SOTA across all 11 datasets. This result also outperforms intensively fine-tuning based medical large language models like Med-PaLM 2 (Singhal et al., 2023b) and Med-Gemini (Saab et al., 2024), establishing a new SOTA on the medical LLM leaderboard. A detailed comparison is shown in Fig. 6.

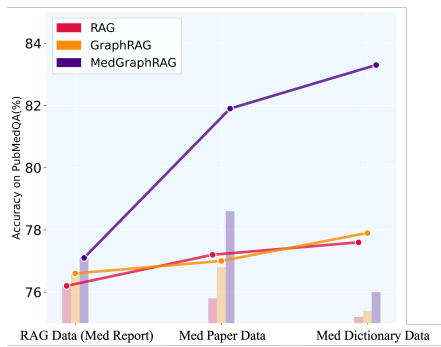


Figure 3: Impact of Repository Data on RAG, GraphRAG, and MedGraphRAG with GPT-4. Line chart: performance with incremental data inclusion; bar chart: performance with individual data inclusion.

3.3.2 Long-form Generation Evaluation

Human Evaluation We conducted human evaluations of long-form model generation on the MultiMedQA and DiverseHealth benchmarks, comparing our method to SOTA models that generate citation-backed responses, including Inline Search in (Gao et al., 2023b), ATTR-FIRST (Slobodkin et al., 2024), and MIRAGE (Qi et al., 2024). Our evaluation panel consisted of 7 certified clinicians and 5 laypersons to ensure feedback from both professional and general users. Raters completed a five-level rating survey for each model’s response, assessing responses across five dimensions: *pertinence* (Pert.), *correctness* (Cor.), *citation precision*

(CP), *citation recall* (CR), and *understandability* (Und.). As shown in Table 2, MedGraphRAG consistently received higher ratings across all metrics. Notably, it showed a significant advantage in CP, CR and Und., indicating that its responses were more often backed by accurate sources and were easier to understand, even for laypersons, thanks to evidence-backed responses and clear explanations of complex medical terms.

Case Study As illustrated in Fig. 7, we compare the responses from GraphRAG and MedGraphRAG for a complex case involving patients with both chronic obstructive pulmonary disease (COPD) and heart failure (left plot). GraphRAG suggested general COPD treatments like bronchodilators and pulmonary rehabilitation but overlooked that certain bronchodilators may worsen heart failure symptoms. In contrast, MedGraphRAG provided a more comprehensive answer by recommending cardioselective beta-blockers—such as bisoprolol or metoprolol—that safely manage both conditions without adverse effects. As we can see from the graph abstracted, this superiority stems from MedGraphRAG’s architecture, where entities are directly linked to key information in references, allowing retrieval of specific evidence. Conversely, GraphRAG struggles to retrieve specific information since its reference and user data are intertwined within the same layer of the graph, which leads to missing key information under the same number of nearest neighbors. And its retrieval based purely on graph summaries results in a lack of detailed insights.

3.4 Ablation and Analysis

3.4.1 Overall Ablation Study

We conducted a comprehensive ablation study to validate the effectiveness of our proposed modules, with the results presented in Table 3. Starting with GraphRAG (Hu et al., 2024) as the baseline, we incrementally incorporated our unique components, including Triple Graph Construction, and U-Retrieval. Notably, both experiments were conducted on the same RAG dataset, eliminating data-related improvements. The results show a gradual performance improvement as more of our modules are added, with significant gains observed when replacing GraphRAG graph construction with our Triple Graph Construction. Additionally, by replacing the summary-based retrieval (Edge et al., 2024b) in GraphRAG with our U-Retrieval method, we

Table 1: Accuracy(%) of LLMs using different retrieval methods. Columns with a blue background represent health fact-checking benchmarks, while the others correspond to medical Q&A benchmarks. The best results are highlighted in bold.

Model	Fake Health	Pub Health	MedQA	Med MCQA	Pub MedQA	MMLU Col-Med	MMLU Col-Bio	MMLU Pro-Med	MMLU Anatomy	MMLU Gene	MMLU Clinic
<i>Baselines without retrieval</i>											
Llama2-13B	53.8	49.4	42.7	37.4	68.0	60.7	69.4	60.3	52.6	66.0	63.8
Llama2-70B	58.9	56.7	43.7	35.0	74.3	64.2	84.7	75.0	62.3	74.0	71.7
Llama3-8B	51.1	53.2	59.8	57.3	75.2	61.9	78.5	70.2	68.9	83.0	74.7
Llama3-70B	64.2	61.0	72.1	65.5	77.5	72.3	92.5	86.7	72.5	83.9	82.7
Gemini-pro	60.6	63.7	59.0	54.8	69.8	69.2	88.0	77.7	66.7	75.8	76.7
GPT-4	71.4	70.9	78.2	72.6	75.3	76.7	95.3	93.8	81.3	90.4	86.2
<i>Baselines with RAG</i>											
Llama2-13B	56.2	54.3	48.1	42.0	68.6	62.5	68.3	63.7	51.0	64.5	67.4
Llama2-70B	64.6	63.2	56.2	49.8	75.2	69.6	85.8	77.4	63.0	75.8	73.3
Llama3-8B	60.5	59.6	64.3	58.2	76.0	68.6	84.9	73.2	72.1	85.2	77.8
Llama3-70B	76.2	72.1	82.3	72.5	80.6	86.8	94.4	89.7	84.3	87.1	87.6
Gemini-pro	72.5	68.4	64.5	57.3	76.9	79.0	91.3	86.4	79.5	80.4	83.9
GPT-4	78.6	77.3	88.1	76.3	77.6	81.2	95.5	94.3	83.1	92.9	93.1
<i>Baselines with GraphRAG</i>											
Llama2-13B	58.7	57.5	52.3	44.6	72.8	64.1	73.0	64.6	52.1	66.2	67.9
Llama2-70B	65.7	63.8	55.1	52.4	74.6	68.0	86.4	79.2	64.6	73.9	75.8
Llama3-8B	61.7	61.0	64.8	58.7	76.6	69.2	84.3	73.9	72.8	85.5	77.4
Llama3-70B	77.7	74.5	84.1	73.2	81.2	87.4	94.8	89.8	85.2	87.9	88.5
Gemini-pro	73.8	70.6	65.1	59.1	75.2	79.8	90.8	85.8	80.7	81.5	84.7
GPT-4	78.4	77.8	88.9	77.2	77.9	82.1	95.1	94.8	82.6	92.5	94.0
<i>Baselines with MedGraphRAG</i>											
Llama2-13B	64.1	61.2	65.5	51.4	73.2	68.4	76.5	67.2	56.0	67.3	69.5
Llama2-70B	69.3	68.6	69.2	58.7	76.0	73.3	88.6	84.5	68.9	76.0	77.3
Llama3-8B	79.9	77.6	74.2	61.6	77.8	89.2	95.4	91.6	85.9	89.3	89.7
Llama3-70B	81.2	79.2	88.4	79.1	83.8	91.4	96.5	93.2	89.8	91.0	94.1
Gemini-pro	79.2	76.4	71.8	62.0	76.2	86.3	92.9	89.7	85.0	87.1	89.3
GPT-4	86.5	83.4	91.3	81.5	83.3	91.5	98.1	95.8	93.2	98.5	96.4

Table 2: Human evaluation on MedQA and Diverse-Health samples.

Data	Methods	Pert.	Cor.	CP	CR	Und.
MultiMedQA	INLINE	91	88	80	74	85
	ATTR.FIRST	93	91	86	77	93
	MIRAGE	95	90	84	75	91
	MedGrapgRAG	97	94	92	86	95
Diverse Health	INLINE	95	84	78	71	81
	ATTR.FIRST	96	91	81	78	85
	MIRAGE	97	89	83	76	87
	MedGrapgRAG	97	96	89	84	93

Table 3: An ablation study of MedGraphRAG, starting from GraphRAG, evaluated using accuracy (%) on Q&A datasets.

	MedQA	PubMedQA	MedMCQA
GraphRAG	88.9	77.9	77.2
+Triple Graph Construction	91.1	81.8	80.9
+U-Retrieval	91.3	83.3	81.5

achieved further improvements, setting new state-of-the-art results across all three benchmarks.

3.4.2 Detailed Ablation on Triple Linking

To assess the individual effects of external RAG data and retrieval technologies, we conducted experiments comparing retrieval methods: RAG, GraphRAG, and MedGraphRAG under two settings: (1) retrieving each tier of data separately (bar

chart in Fig. 3), and (2) incrementally adding all three tiers (line chart in Fig. 3). The results show that both the data and the right retrieval method must work together to unlock the full potential. When retrieving data by standard RAG, Med-Paper data individually improves performance by less than 2%, and Med-Dictionary data by less than 1%. Accumulating three tier data also leads to mediocre improvements. GraphRAG shows improvement in retrieving individual data but has minimal gains when incrementally adding more data, likely due to superficiality from linking trivial entities, as discussed in the previous case study. In contrast, Med-GraphRAG efficiently handles the additional data, using its hierarchical structure to clarify relationships and show strong improvements as more data is added. With MedGraphRAG, we see significant improvements of over 6% and 8% for Med-Paper and Med-Dictionary data, respectively, highlighting the importance of the retrieval method in maximizing the impact of the data.

3.5 Detailed Ablation on U-Retrieval

In U-Retrieval, we set the retrieval depth to 4-5 layers, the number of retrieval entities to 60, and entity neighbors to 16. These settings were de-

terminated through comprehensive trials. First, we examine the impact of the retrieval range, i.e. the number of entities and neighbors, using GPT-4 with MedGraphRAG on MedQA, as shown in Fig. 4. Our findings show that retrieving more data does not necessarily lead to better performance. In fact, more data can introduce noise and exacerbate LLM performance issues with long contexts. The peak performance occurs when the retrieval size reaches approximately 120 entities with 4-hop neighbors or 60 entities with 16-hop neighbors. The 16-hop neighbors setting performed slightly better, likely due to the robustness of graph-based linking compared to vector-similarity-based retrieval.

As previously mentioned, there is also a trade-off between model accuracy and response time with retrieval layer increases. This relationship is explored in Fig. 5. The figure compares the cost time and MedQA accuracy across retrieval depths from 0 to 9 layers. We observe that both performance and response time increase as the retrieval layer increases initially. However, performance begins to degrade when retrieving more layers, as higher layers often contain less relevant information, which can interfere with refining the response. The optimal retrieval depth is 4 layers for the GPT-4 model and 5 layers for others, which we use as the default setting in our experiments.

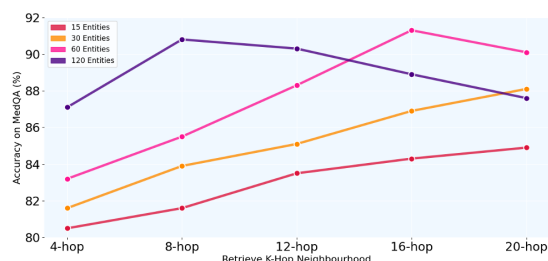


Figure 4: The effect of retrieving different number of entities and neighbourhods. Performance evaluated by GPT-4 (MedGraphRAG) on MedQA.

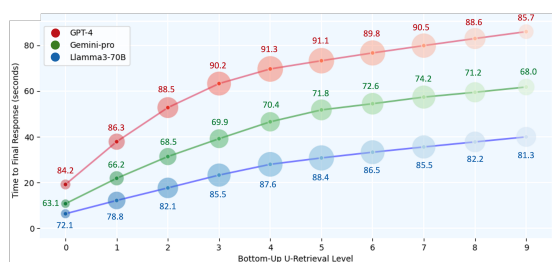


Figure 5: The relationship between U-retrieval level and time cost.

4 Related Work

Large language models (LLMs) built on Transformer architectures have advanced rapidly, leading to specialized medical LLMs such as BioGPT (Luo et al., 2022), PMC-LLaMA (Wu et al., 2023), BioMedLM (Bolton et al., 2022), and Med-PaLM 2 (Singhal et al., 2023b). While many are fine-tuned by large organizations, recent research has focused on cost-efficient, non-fine-tuned approaches, primarily using prompt engineering (Saab et al., 2024; Wang et al., 2023; Savage et al., 2024). RAG, as another non-finetuning approach, is rarely explored for medical applications (Miao et al., 2024; Xiong et al., 2024; Long et al., 2024) and lacks support for evidence-based responses and term explanations required in clinical settings.

RAG (Lewis et al., 2021) enables models to use specific datasets without additional training, improving response accuracy and reducing hallucinations (Gua et al., 2020). RAG has shown strong results across various tasks, including generating responses with citations (Gao et al., 2023b; Slobodkin et al., 2024; Qi et al., 2024; Nakano et al., 2021; Bohnet et al., 2022; Gao et al., 2023a,c; Schimanski et al., 2024; Zhang et al., 2024). GraphRAG (Hu et al., 2024) further enhances complex reasoning by constructing knowledge graphs, but lacks specific design features for generating attributed responses, and its application in medical specialization remains limited.

5 Conclusion

MedGraphRAG improves the reliability of medical response generation with its graph-based RAG framework, using Triple Graph Construction and U-Retrieval to enhance evidence-based, context-aware responses. Future work will focus on real-time data updates and validation on real-world clinical data.

6 Limitation

Despite the strong capabilities demonstrated by MedGraphRAG, the graph construction step incurs significant computational costs. In the retrieval and response stage, although the costs are lower than graph construction, they remain higher than standard large language model (LLM) calls, with each question taking around 70 seconds to process (see Figure 6 for details). Future efforts should explore methods to transfer pre-constructed graphs or accelerate the graph construction process to mitigate these computational costs.

Additionally, the scale of experimental data and the expensive nature of graph construction make it challenging to conduct comprehensive comparisons of hyper-parameter settings and technology choices. For instance, factors such as the number of paragraphs in the context window during document chunking, the use of alternative RAG datasets, and the impact of different prompts for graph construction were selected empirically based on limited data. A more rigorous and comprehensive comparison of these factors is needed in future work to identify the optimal configurations that maximize the method’s potential.

For latency, while our method introduces additional computational overhead, we believe that in critical fields like medicine, users are often willing to trade speed for precision. As demonstrated in our manuscript, our approach delivers significantly more accurate and evidence-based responses. A useful analogy is the increasing popularity of GPT-based deep research assistants, which users accept despite longer response times in exchange for higher-quality, more professional outputs. On the graph updating side, we designed the graph structure with hierarchical modularity to accommodate different update frequencies: The bottom layer contains foundational medical dictionaries and terminology, which change infrequently and can be treated as static. The middle layer integrates moderately updated sources like medical literature. The top layer includes frequently changing sources such as clinical reports. Since updates to the lower layers are more costly while the upper layers are more lightweight and cost-efficient to update, the differing update frequencies across layers naturally align with this structure—thereby helping to reduce the overall update cost to some extent. In the future work, to address the remaining challenge of expensive updates even at the top layer, we can propose

a local update strategy. Specifically, we can compute the semantic distance between newly inserted knowledge and existing Meta-Graphs, and apply updates only to relevant subgraphs that exceed a defined threshold. This selective updating approach balances both efficiency and accuracy. We recognize these as practical and important limitations, and we plan to supply more detailed discussion on them as part of our future work in this research direction.

Finally, regarding human evaluation, while we made efforts to ensure diversity and expertise among our raters, the evaluation may still carry biases due to the limited sample size (120 questions on MultiMedQA and 50 questions on DiverseHealth). Future research should include larger-scale and better-designed human evaluations to thoroughly assess the model’s performance.

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A Detailed Implementation

In the semantic document chunking process, we apply proposition transfer (Chen et al., 2023) to each paragraph before semantic validation to extract standalone statements that are self-contained and unambiguous (e.g., transforming "It prevents respiratory disease" to "Remdesivir prevents respiratory disease"). Through proposition transfer, each paragraph is transformed into independent, clear statements. For semantic validation, we utilize an LLM to first generate a short summary and a title for the current chunk. The LLM then determines if the current paragraph belongs to this chunk based on the title and summary. If it belongs, the LLM updates the title and summary accordingly. If not, the current chunk is finalized, and the LLM generates a title and summary for the new paragraph, treating it as the start of a new chunk. If the scan reaches the end of the document, the current chunk is automatically finalized to ensure no chunk spans across multiple documents.

In the entity extraction, we include unique IDs to trace their source document. In practice, for the user privacy data, we generate a universally unique identifier (UUID) for each document as their IDs. For the medical papers and books, we use their Digital object identifier (DOI) as their IDs, and for the medical dictionaries, we use their UMLS Concept Unique Identifiers (CUI) as their IDs. This identifier is crucial for retrieving information from the source, enabling the generation of evidence-based responses later. For tag-based summary generation and merging, we insert ten tags into the prompt at a time to iteratively generate the response.

For the standard LangChain RAG baseline, we followed the official implementation, which uses similarity search based on cosine similarity between the embedded query and the embedded documents in a vector store. In our experiments, we used this default setup to ensure a fair and reproducible comparison. Since in MedGraphRAG, besides the contributed U-Retrieval, we also relied on cosine similarity to retrieve the final bottom-level Meta-Graph. This design choice ensures consistency across all baselines and isolates the impact of our proposed retrieval strategy.

For testing the models on MultiMedQA, we evaluate their zero-shot performance using only the test set of each dataset, without utilizing the training data for fine-tuning or including it in the RAG data for retrieval. For evaluating accuracy on FakeHealth, we incorporate its news content into the Medical-Papers-tier graph of MedGraphRAG and into RAG data of the others, then use the criteria questions from the news content to prompt the models to respond with 'Satisfactory' or 'Not Satisfactory.' For PubHealth, we integrate its news/reviews into Medical-Papers-tier graph of MedGraphRAG and into RAG data of the others, and prompt the models to classify each claim as 'True,' 'False,' 'Unproven,' or a 'Mixture.'

B Additional Results and Analysis

B.1 Compare to SOTA Medical LLM Models

We also evaluated MedGraphRAG against a range of previous SOTA medical large language models on these benchmarks, including both intensively fine-tuned models (Gu et al., 2022)(Yasunaga et al., 2022a)(Yasunaga et al., 2022b)(Bolton et al., 2022)(Singhal et al., 2022)(Singhal et al., 2023a)(Wu et al., 2023) and non-fine-tuned models (Nori et al., 2023)(OpenAI, 2023b)(OpenAI, 2023a)(Saab et al., 2024). The results, depicted in Fig. 6, show that when combined with GPT-4, our MedGraphRAG surpasses the previous SOTA model, Medprompt (Nori et al., 2023), by a notable 1.1% on the MedQA benchmark, and also outperforms it across all 9 datasets, establishing a new SOTA on the medical LLM leaderboard. It's important to note that while Medprompt retrieves training data with similar questions and correct answers as examples for prompting, our model operates with a simple prompt containing only the original question. This improvement further demonstrates MedGraphRAG's superior capa-

Table 4: Compare to several specialized medical retrievers across nine medical Q&A benchmarks.

	MedQA	MedMCQA	PubMedQA	MMLU-Col-Med	MMLU-Col-Bio	MMLU-Pro-Med	MMLU-Anatomy	MMLU-Gene	MMLU-Clinic
MedCPT	79.6	74.9	76.8	77.8	95.4	93.9	82.6	90.9	88.3
MedRAG	88.5	78.1	78.9	85.5	96.8	94.8	84.5	93.6	94.5
RAG2	85.2	76.2	79.3	83.4	96.1	94.8	83.9	91.0	93.2
Self-BioRAG	81.1	73.5	76.2	84.1	95.7	94.2	82.1	92.8	92.7
Ours	91.3	81.5	83.8	91.5	98.1	95.8	93.2	98.5	96.4

bility, even when retrieving from data with a different distribution. Furthermore, when compared to intensive fine-tuning methods on these medical datasets, MedGraphRAG outperforms strong models like Med-PaLM 2 (Singhal et al., 2023b) and Med-Gemini (Saab et al., 2024), establishing a new SOTA. This superior performance highlights MedGraphRAG’s ability to efficiently leverage the inherent capabilities of LLMs and enhance their performance with additional data, without the need for fine-tuning.

B.2 Case study: GPT4 with and without MedGraphRAG

As shown in Fig. 7, we compare the responses generated by vanilla GPT-4 and MedGraphRAG for a misleading case where a patient presents with symptoms commonly associated with Alzheimer’s but is actually Vascular Dementia. GPT-4 was misled, returning an incorrect diagnosis. In contrast, MedGraphRAG notes the details like that the MRI showed moderate vascular changes and white matter lesions, which are indicative of chronic ischemic damage—typical of vascular dementia rather than Alzheimer’s, through retrieving the findings in (Smith and Beaudin, 2018), "CBF and WMH that...causing ical impairments,". With detailed definitions of medical terms and source knowledge retrieved to assist the reasoning process, MedGraphRAG chose the correct answer and provided a detailed, easily understandable explanation with citation, enabling users to verify the response.

In Table 4, we compared MedGraphRAG with several specialized medical retrievers, including MedCPT (Jin et al., 2023), MedRAG (Zhao et al., 2025), RAG2 (Sohn et al., 2024), and Self-BioRAG (Jeong et al., 2024), across nine medical Q&A benchmarks. All methods were evaluated under the same RAG corpus and experimental settings, as described in the manuscript. The results show that MedGraphRAG consistently outperforms all other specialized retrievers across all datasets, with significant performance gains. We attribute this improvement to our method’s ability to semantically organize large-scale medical corpora, enabling pre-

cise and context-aware retrieval even in complex and long-range RAG corpora.

B.3 Case study: Long-form generation of MedGraphRAG

We provided four examples of MedGraphRAG Long-form response generation. We include the diverse cases across Comorbidity Fig. 8, 9, Rare Disease Fig. 10,11, Minority Health Fig. 12,13, and Chronic Disease Management Fig. 14,15. We can see the unique responses provided by MedGraphRAG combining citations with clear term explanations in medical responses ensures both credibility and understanding. Citations provide a foundation of evidence, reassuring patients and professionals that recommendations are grounded in research. For example, in the hormone replacement therapy answer, the association between HRT and increased risks of cardiovascular events and thromboembolic complications is backed by "Dhejne et al., 2011," which provides long-term follow-up data on health outcomes in transgender individuals undergoing hormone therapy. This level of transparency is particularly important in healthcare, where trust is critical for patient compliance and effective care.

Clear term explanations help bridge the gap for those who might struggle with medical jargon. By explaining complex terms like cardioselective beta-blockers or hypoglycemia in simple language, patients better understand their condition and the rationale behind their treatment. This not only empowers them but also helps in preventing misunderstandings that could lead to improper management of their health. Altogether, using citations for evidence and plain language for explanation strikes the right balance between trust, safety, and accessibility in medical communication.

B.4 Case study: Abstracted Graph comparison between GraphRAG and MedGraphRAG

We conducted a closer examination of the abstracted graphs of GraphRAG (Fig. 16 a) and MedGraphRAG (Fig. 16 b) for the case study shown in the left plot of Fig. 7. By abstracting similar nearest neighbors of the retrieved entities (COPD and

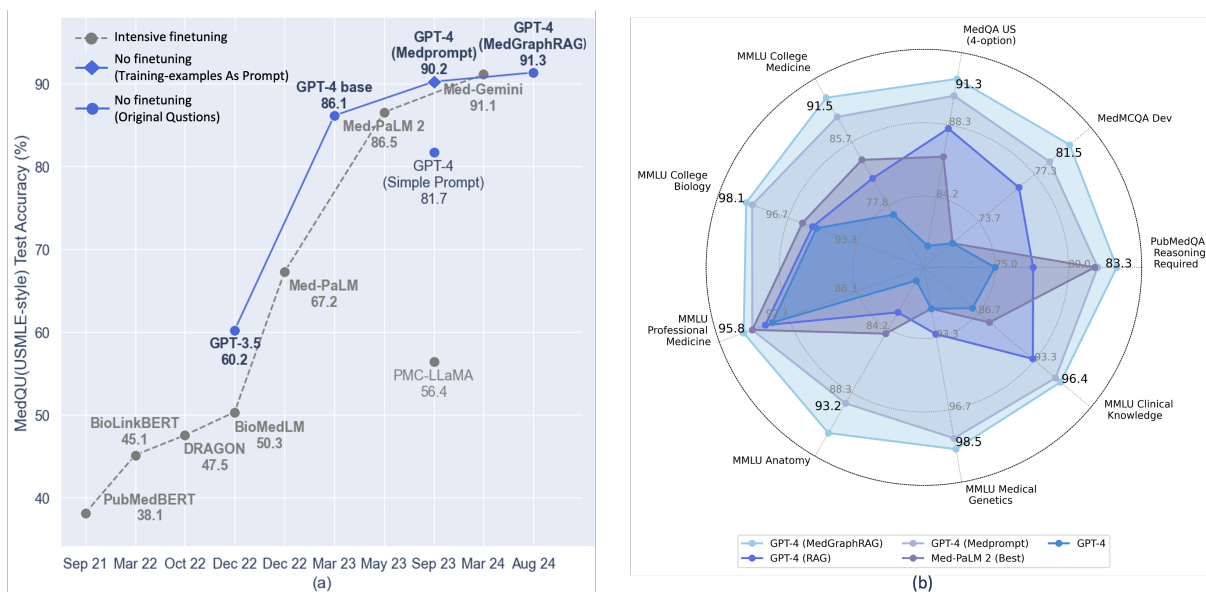


Figure 6: Compare to SOTA Medical LLM Models on MedQA benchmark.

Heart Failure), we observed that MedGraphRAG accessed more detailed and specific entities, such as beta-1 receptors and Cardioselective Beta-Blockers, by linking to relevant references. While these entities are also present in the GraphRAG graph, they were not retrieved under the same number of nearest neighbors due to their indirect linkage with the retrieved entities. GraphRAG lacks a hierarchical graph that directly links these entities through an "is reference of" relationship, leading them to be overshadowed by more general neighbors at the same tier, ultimately missing retrieval.

Moreover, MedGraphRAG’s approach to linking Heart Failure with Cardioselective Beta-Blockers enables further connections through beta-1 receptors in the second-tier graph, eventually linking back to Non-selective Beta-Blockers. It helps to link Heart Failure and Non-selective Beta-Blockers as neighbors in the first-tier graph relationship linking stage, which significantly enhances the LLM’s ability to generate specific and accurate responses. Such an observation demonstrates the importance of including triple linking relationships when constructing the first-tier graph. MedGraphRAG leverages this unique design to build a more detailed and professional knowledge graph, resulting in better entity retrieval and richer context for accurate generation.

C Boarder Impact

Our MedGraphRAG enhances LLMs by providing intrinsic source citations, significantly improv-

ing transparency, interpretability, and verifiability. This tool ensures that each response generated by the LLM is accompanied by provenance or source grounding information, clearly demonstrating that answers are rooted in the dataset. The availability of cited sources for each assertion allows users, especially in the critical field of medicine, to swiftly and accurately audit the LLM’s output against the original source material. This feature is crucial where (a) a trustworthy relationship -based on transparency of reasoning, needs to exist between human experts and LLM agents and (b) where the security/privacy of the information shared with the agent is mission critical. By employing this method, we have developed an Evidence-based Medical LLM. Clinicians can easily verify the source of the reasoning and adjust the model’s responses, ensuring the safe use of LLMs in clinical scenarios.

Question

Question:

Mr. John Smith, a 68-year-old retired engineer, presents with increasing forgetfulness and occasional episodes of sudden confusion and speech difficulty. His medical history includes type 2 diabetes, hypertension, and a transient ischemic attack two years ago. Neuroimaging reveals moderate vascular changes and white matter lesions, while his PET scan shows patchy decreased glucose metabolism. Given these symptoms and findings, which of the following is the most likely diagnosis?

Options:

- A. Alzheimer's Disease
- B. Vascular Dementia
- C. Frontotemporal Dementia
- D. Lewy Body Dementia

Answer:

B. Vascular Dementia

GPT-4

Answer:

A. Alzheimer's Disease

Explanation:

1.Cognitive Testing: His moderate cognitive impairment on the Montreal Cognitive Assessment (MoCA) might also be consistent with Alzheimer's, considering the general decline in cognitive functions.

2.Neuroimaging: Although the MRI shows vascular changes, the patchy areas of decreased glucose metabolism observed in the PET scan can sometimes overlap with early Alzheimer's patterns, reflecting disrupted brain activity.

3.Symptom Pattern: While his symptoms show vascular influences, the general forgetfulness and worsening cognitive decline over the past year could also align with the progressive nature of Alzheimer's disease.

MedGraphRAG

Answer:

B. **Vascular Dementia**

Explanation:

Medical History and Symptoms: His history of vascular risk factors and the nature of his **episodic symptoms**, which include sudden confusion and difficulties with speech, are indicative of neurological disturbances typically associated with vascular incidents like **micro-strokes or TIAs** [1].

1.Neuroimaging Results: The MRI showing moderate vascular changes and **white matter lesions** is characteristic of **chronic ischemic damage**, prevalent in vascular dementia, rather than the **neurodegenerative patterns** typically observed in Alzheimer's or other dementias [2].

2.Additional Findings: His **episodic cognitive symptoms** also support a vascular origin rather than a degenerative neurological disease like Alzheimer's, which would generally present a gradual, consistent cognitive decline rather than episodic [3].

These factors collectively support the diagnosis of vascular dementia, aligning more with the implications of his vascular medical history and the episodic nature of his cognitive disturbances.

Vascular Dementia: A form of dementia caused by an impaired supply of blood to the brain, often resulting from strokes or other vascular injuries, which leads to cognitive decline.

Episodic symptoms: Symptoms that occur in discrete episodes, rather than being continuous. These can vary in severity and nature and are transient, typically related to neurological events.

Micro-strokes or TIAs: Micro-strokes are small, unnoticed strokes that can cause temporary, minor neurological symptoms. TIAs are brief episodes of neurological dysfunction caused by loss of blood flow to the brain, typically lasting less than 24 hours without causing permanent damage.

White matter lesions: These are indicators of vascular pathology in the brain, including changes in blood vessel structure and areas of damage in the white matter, often due to chronic ischemic (lack of blood flow) conditions.

Chronic ischemic damage: Long-standing damage to brain tissue caused by repeated or persistent reduction in blood supply, typically resulting in loss of neuronal function and health.

Neurodegenerative patterns: Refers to the structural and functional changes in the brain associated with progressive neurological diseases like Alzheimer's, where there is gradual degradation and death of neurons.

Episodic cognitive symptoms: Fluctuating cognitive impairments that appear in episodes, contrasted with steady cognitive decline seen in degenerative diseases.

[1]: O'Brien, J. T., & Thomas, A. (2015). Vascular dementia. *Lancet*, 386(10004), 1698-1706.

[2]: Smith, E. E., & Beaudin, A. E. (2018). New insights into cerebral small vessel disease and vascular cognitive impairment from MRI. *Current Opinion in Neurology*, 31(1), 36-43.

[3]: Gorelick, P. B., Scuteri, A., Black, S. E., et al. (2011). Vascular contributions to cognitive impairment and dementia: A statement for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*, 42(9), 2672-2713.

Figure 7: Example case shows MedGraphRAG generating evidence-based responses with grounded citations and terminology explanations.

Comorbidity

Question

In patients with both chronic obstructive pulmonary disease (COPD) and heart failure, how can lung function be improved?

MedGraphRAG

In patients with both **COPD** and **heart failure**, the combination of **bronchodilators** and **cardioselective beta-blockers** can be used effectively to manage both conditions without significantly worsening respiratory symptoms or heart function [1,4]. **Cardioselective beta-blockers**, such as bisoprolol or metoprolol, are safe for **COPD** patients because they primarily block **beta-1 receptors** in the heart, minimizing the risk of **bronchoconstriction** associated with non-selective beta-blockers [1,2]. **Bronchodilator** therapy, including long-acting beta-2 agonists and **muscarinic antagonists**, should be continued to improve lung function in **COPD** [3].

Bronchodilators, such as long-acting beta-2 agonists (LABAs), help relax the airway muscles, which reduces **bronchoconstriction** and increases airflow, thereby improving lung function [3,5]. **Muscarinic antagonists**, like tiotropium, work by blocking the action of acetylcholine on airway muscles, further aiding in bronchodilation and easing the symptoms of **COPD** [5]. These medications should be continued as part of the treatment regimen to enhance respiratory efficiency.

Cardioselective beta-blockers, such as bisoprolol and metoprolol, are particularly beneficial for **COPD** patients as they predominantly target **beta-1 receptors** in the heart, reducing cardiovascular strain without significantly affecting the **beta-2 receptors** that are crucial for lung function [1,2]. This selective targeting helps avoid the **bronchoconstriction** risk often seen with non-selective **beta-blockers**, which can negatively impact patients with respiratory conditions [2,6].

For patients managing both **COPD** and **heart failure**, the use of **cardioselective beta-blockers** in combination with **bronchodilators** can be a balanced and effective strategy to optimize both cardiovascular and respiratory health. Studies have shown that the careful use of **cardioselective beta-blockers** reduces mortality risk without exacerbating respiratory symptoms [2]. This combination treatment helps maintain heart function while avoiding respiratory compromise, ultimately contributing to improved lung function and better overall outcomes in patients with these comorbid conditions.

References:

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2. Du Q, Sun Y, Ding N, Lu L. "Beta-blockers reduced the risk of mortality and exacerbation in patients with COPD: A meta-analysis of observational studies." *PLoS ONE*. 2016; 11(11): e0166861.
3. O'Donnell DE, Parker CM. "COPD exacerbations . 3: Pathophysiology." *Thorax*. 2006.
4. Hawkins NM, Petrie MC, Jhund PS, Chalmers GW, Dunn FG, McMurray JJ. "Heart failure and chronic obstructive pulmonary disease: diagnostic pitfalls and epidemiology." *European Journal of Heart Failure*. 2009; 11(2): 130-139.
5. Tashkin DP, Celli B, Senn S, et al. "A 4-year trial of tiotropium in chronic obstructive pulmonary disease." *New England Journal of Medicine*. 2008; 359(15): 1543-1554.
6. Rutten FH, Cramer MJ, Lammers JW, Grobbee DE, Hoes AW. "Heart failure and chronic obstructive pulmonary disease: An ignored combination?" *European Journal of Heart Failure*. 2006; 8(7): 706-711.

Figure 8: MedGraphRAG long-form generation response on Comorbidity example-part1.

Terminological Clarification:

•**COPD (Chronic Obstructive Pulmonary Disease):** A chronic inflammatory lung disease that obstructs airflow, causing breathing difficulties, cough, and mucus production.

•**Heart Failure:** A condition in which the heart cannot pump blood effectively, leading to symptoms such as fatigue, shortness of breath, and fluid retention.

•**Bronchodilators:** Medications that relax the muscles surrounding the airways, leading to widened air passages and improved airflow to ease breathing in **COPD** patients.

•**Cardioselective Beta-Blockers:** A type of **beta-blocker** that primarily targets **beta-1 receptors** in the heart, which reduces the risk of respiratory complications. They are used to manage heart conditions without significantly affecting lung function.

•**Beta-Blockers:** A class of medications used to manage cardiovascular conditions like hypertension and **heart failure**. They work by blocking **beta receptors** to decrease heart rate and blood pressure. Non-selective **beta-blockers** can also affect **beta-2 receptors**, leading to **bronchoconstriction**.

•**Beta-1 Receptors:** Receptors primarily found in the heart that, when blocked, lead to reduced heart rate and force of contraction, making them a key target for treating cardiovascular diseases.

•**Beta-2 Receptors:** Receptors found in the smooth muscles of the airways. Blocking these receptors can lead to **bronchoconstriction**, which is why cardioselective **beta-blockers** that do not significantly impact these receptors are preferred for **COPD** patients.

•**Bronchoconstriction:** The narrowing of the airways in the lungs due to the tightening of surrounding muscles, leading to difficulty in breathing. This is a concern for **COPD** patients and can be exacerbated by non-selective **beta-blockers**.

•**Beta-2 Agonists:** Medications that stimulate **beta-2 receptors** to relax airway muscles and promote bronchodilation, improving airflow in **COPD** patients.

•**Muscarinic Antagonists:** Medications that block the action of acetylcholine on muscarinic receptors, which helps to relax the airways and improve airflow, thus reducing symptoms in **COPD**.

Figure 9: MedGraphRAG long-form generation response on Comorbidity example-part2.

Question

Rare Disease

What are the most effective strategies for managing progressive thoracic insufficiency in patients with fibrodysplasia ossificans progressiva (FOP)?

MedGraphRAG

Managing progressive **thoracic insufficiency** in **FOP** patients requires a careful balance between effective respiratory support and preventing **heterotopic ossification (HO)** [2]. Early use of **non-invasive ventilation (NIV)** [1], such as **BiPAP**, particularly during sleep or in response to **respiratory infections**, can improve respiratory function without invasive procedures [2], minimizing the risk of trauma [3]. Gentle, supervised **physical therapy** helps maintain lung capacity while avoiding trauma that could trigger **HO** [4].

Non-invasive ventilation (NIV), like **BiPAP**, can be used to provide respiratory support, especially during periods of compromised breathing due to **respiratory infections** or sleep. This avoids the need for invasive procedures, which can provoke **HO** in **FOP** patients [1,2]. **BiPAP** specifically offers different levels of air pressure during inhalation and exhalation, making it a comfortable and effective choice for improving respiratory function [3,5].

Supervised **physical therapy** plays a crucial role in maintaining lung capacity while minimizing trauma. It is important that such exercises are gentle and performed under supervision to avoid any injury that might lead to the development of **heterotopic ossification** [4,6]. The main goal is to preserve thoracic movement without stimulating the abnormal bone growth characteristic of **FOP**. **Nutritional support** is also essential to optimize respiratory function in **FOP** patients. Maintaining adequate nutrition helps to improve overall health and supports the muscles needed for breathing [7].

References:

- 1.ShulmanI, et al. "Maternal genetic skeletal disorders: lessons learned from cases of maternal osteogenesis imperfecta and fibrodysplasia ossificans progressiva." *Journal of Clinical*.
- 2.Pignolo RJ, et al. "The natural history of flare-ups in fibrodysplasia ossificans progressiva (FOP): a comprehensive global assessment." *Journal of Bone and Mineral Research*. 2016.
- 3.Kaplan FS, et al. "Fibrodysplasia ossificans progressiva: Clinical and genetic aspects." *Orphanet Journal of Rare Diseases*. 2008.
- 4.Additional source not explicitly listed (general reference to supervised physical therapy for maintaining lung capacity in **FOP** patients).
- 5.Rocke DM, et al. "Pulmonary function and its management in FOP." *Respiratory Care*. 2017.
- 6.Kaplan FS, Glaser DL, Shore EM, et al. "The medical management of fibrodysplasia ossificans progressiva: current treatment considerations." *Clinical Reviews in Bone and Mineral Metabolism*. 2005.
- 7.Al Mukaddam M, et al. "Nutrition in Fibrodysplasia Ossificans Progressiva: Strategies and Management." *Journal of Clinical Endocrinology & Metabolism*. 2020.

Figure 10: MedGraphRAG long-form generation response on Minority Health example-part1.

Terminological Clarification:

- Heterotopic Ossification (HO):** The formation of bone tissue in areas outside of the skeletal system, usually in soft tissues like muscles. It is common in **FOP** and can be triggered by trauma or injury, resulting in limited mobility and pain.
- Non-Invasive Ventilation (NIV):** A respiratory support method that uses a mask to deliver air to the lungs, assisting breathing without requiring surgical procedures. It is commonly used to avoid complications in **FOP** patients where trauma can lead to **HO**.
- BiPAP (Bilevel Positive Airway Pressure):** A type of **NIV** that provides two levels of air pressure—higher pressure when inhaling and lower pressure when exhaling. It helps maintain effective ventilation and supports breathing during sleep or when respiratory function is compromised.
- Fibrodysplasia Ossificans Progressiva (FOP):** A rare genetic disorder characterized by the formation of bone in muscles, tendons, and other connective tissues, leading to progressive immobility. Even minor trauma can trigger flare-ups, resulting in abnormal bone growth.
- Thoracic Insufficiency:** A condition where the chest wall cannot adequately support normal breathing or lung growth, leading to reduced lung function. It is a common complication in **FOP** patients due to **HO** affecting the ribcage and spine.
- Respiratory Infections:** Infections affecting the respiratory tract, which can significantly worsen breathing in **FOP** patients. Prevention and effective management are crucial to avoid triggering **HO**.
- Physical Therapy:** Supervised exercises designed to maintain or improve mobility. For **FOP** patients, it must be done cautiously to avoid any physical stress or trauma that could lead to **HO**.
- Nutritional Support:** Adequate nutrition is crucial for maintaining general health and supporting respiratory muscle function, especially in **FOP** patients where compromised lung capacity can lead to difficulties in meeting nutritional needs.

Figure 11: MedGraphRAG long-form generation response on Minority Health example-part2.

Minority Health

Question

Will long-term hormone replacement therapy (HRT) injections affect life expectancy in a male-to-female transgender individual?

MedGraphRAG

Yes, long-term **hormone replacement therapy (HRT)** injections in **male-to-female transgender** individuals can affect life expectancy by potentially increasing certain health risks [1,4], particularly **cardiovascular events** [2] and **thromboembolic complications** [3]. However, with proper medical supervision [5], **risk assessment**, and **management strategies**, these risks can be mitigated, leading to a life expectancy similar to the general population.

Proper medical supervision is crucial for monitoring hormone levels, screening for potential complications, and managing any risks associated with long-term **HRT** [5]. **Cardiovascular events**, such as heart attacks or strokes, are potential risks that should be closely monitored in individuals undergoing long-term **HRT**, especially when risk factors like smoking or pre-existing heart conditions are present [2].

Thromboembolic complications, including deep vein thrombosis (DVT) or pulmonary embolism, are also possible but can be managed through careful risk stratification and preventive measures [3,6]. With regular medical check-ups and the use of personalized **management strategies**, individuals undergoing **HRT** can mitigate these risks and achieve a life expectancy comparable to that of the general population.

References:

- 1.Dhejne C, et al. "Long-Term Follow-Up of Transsexual Persons Undergoing Sex Reassignment Surgery: Cohort Study in Sweden." *PLoS ONE*. 2011.
- 2.Nota NM, et al. "Occurrence of Acute Cardiovascular Events in Transgender Individuals Receiving Hormone Therapy." *European Journal of Endocrinology*. 2019.
- 3.Asscheman H, et al. "Venous Thrombosis and Long-Term Use of Estrogens in Transgender Women." *Journal of Sexual Medicine*. 2014.
- 4.Wierckx K, et al. "Long-term evaluation of cross-sex hormone treatment in transsexual persons." *Journal of Clinical Endocrinology & Metabolism*. 2012.
- 5.Hembree WC, et al. "Endocrine Treatment of Gender-Dysphoric/Gender-Incongruent Persons: An Endocrine Society Clinical Practice Guideline." *Journal of Clinical Endocrinology & Metabolism*. 2017.
- 6.Seal LJ, et al. "Health Implications of Gender Affirmation in Transgender Individuals." *Lancet Diabetes Endocrinology*. 2016.

Figure 12: MedGraphRAG long-form generation response on Rare Disease example1.

Terminological Clarification:

•**Hormone Replacement Therapy (HRT):** A treatment involving the administration of hormones, such as estrogen or testosterone, to align an individual's physical characteristics with their gender identity. In **male-to-female transgender** individuals, **HRT** typically includes estrogen and anti-androgens.

•**Male-to-Female Transgender:** Refers to individuals assigned male at birth who identify and transition to female. **HRT** is a part of the transition process that helps in developing feminine physical traits.

•**Cardiovascular Events:** Medical conditions affecting the heart or blood vessels, including heart attacks, strokes, and other complications related to the cardiovascular system. These events can be a risk factor for individuals on long-term **HRT**.

•**Thromboembolic Complications:** Conditions that involve blood clots forming in the veins, which may lead to **deep vein thrombosis (DVT)** or **pulmonary embolism**. These are possible risks associated with **HRT** in transgender individuals.

•**Proper Medical Supervision:** Regular healthcare support that includes monitoring hormone levels, evaluating risk factors, and adjusting treatments as necessary. This helps mitigate potential risks associated with **HRT**.

•**Risk Assessment:** The process of identifying potential health risks, such as cardiovascular issues or thromboembolic complications, in individuals undergoing **HRT**.

•**Management Strategies:** Techniques or treatments designed to minimize health risks. For individuals on **HRT**, this may include lifestyle modifications, regular monitoring, and tailored medical care to address potential complications.

•**Deep Vein Thrombosis (DVT):** A condition where a blood clot forms in a deep vein, usually in the legs. **DVT** is one of the possible **thromboembolic complications** associated with **HRT**.

•**Pulmonary Embolism:** A blockage in one of the pulmonary arteries in the lungs, usually caused by a blood clot that has traveled from the legs (**DVT**). It is a serious complication that can occur with long-term **HRT**.

Figure 13: MedGraphRAG long-form generation response on Rare Disease example-part2.

Question

Chronic Disease Management

I have Type 1 Diabetes and I'm planning to exercise for about 30 minutes after work. How should I adjust my insulin and eating?

MedGraphRAG

To avoid **low blood sugar** during and after your 30-minute exercise session, start by checking your **blood glucose** before you begin; if it's below your target range, consume a small **carbohydrate** snack (about 15–30 grams) [2,3,6]. Consider reducing your **rapid-acting insulin** dose before exercise by 25–50% if you've taken it within the last few hours [2,3,7], as exercise increases **insulin sensitivity** and can lower **blood sugar levels** [1,2,3,4]. Keep **fast-acting carbohydrates** readily available during your workout in case you experience symptoms of **hypoglycemia** [1,3,5].

Monitoring **blood glucose** levels before, during, and after exercise helps maintain stable **blood sugar levels**. Consuming **carbohydrates** before exercise if **blood glucose** is low will help prevent **hypoglycemia** and allow you to exercise more comfortably [2,3,6]. **Rapid-acting insulin** adjustments are important, as exercise can significantly increase **insulin sensitivity**, leading to a faster decline in **blood sugar levels** [2,3,7].

Fast-acting carbohydrates (such as glucose tablets or sugary drinks) should be easily accessible during exercise in case of **hypoglycemia**. These types of carbohydrates act quickly to raise **blood sugar levels** and can help stabilize levels if they drop too low during activity [1,2,3,5].

References:

1. Bakris, G., et al. "American Diabetes Association (ADA) standards of medical care in diabetes 2016 cardiovascular disease and risk management." *Diabetes Care*. 2016.
2. Riddell MC, et al. "Exercise management in type 1 diabetes: a consensus statement." *The Lancet Diabetes & Endocrinology*. 2017.
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6. Rabasa-Lhoret R, et al. "Exercise in Type 1 Diabetes: A practical review of its benefits and challenges." *Journal of Diabetes and its Complications*. 2009.
7. Heinemann L, et al. "Adjustment of insulin therapy for physical activity in type 1 diabetes mellitus." *Diabetes Obesity and Metabolism*. 2014.

Figure 14: MedGraphRAG long-form generation response on Chronic Disease Management example-part1.

Terminological Clarification:

•**Low Blood Sugar (Hypoglycemia):** A condition where **blood glucose** levels fall below the normal range, leading to symptoms like shakiness, sweating, confusion, or even loss of consciousness if untreated.

•**Blood Glucose:** The concentration of glucose (sugar) in the blood, commonly referred to as **blood sugar levels**. Monitoring **blood glucose** is crucial for individuals with **Type 1 Diabetes** to manage their health.

•**Carbohydrate:** A macronutrient found in foods such as bread, fruits, and sweets that provides energy. Consuming **carbohydrates** before exercise can help maintain **blood glucose** levels, especially in individuals with diabetes.

•**Rapid-Acting Insulin:** A type of insulin that starts to work quickly to reduce **blood glucose** levels, typically within 15 minutes of injection. It helps manage the spikes in **blood sugar** that occur after meals.

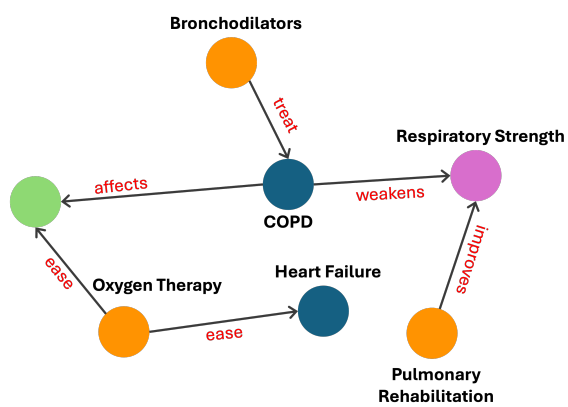
•**Insulin Sensitivity:** The body's responsiveness to insulin, meaning how effectively insulin helps cells absorb glucose. Exercise increases **insulin sensitivity**, which means the body requires less insulin to lower **blood glucose** levels.

•**Blood Sugar Levels:** The amount of glucose present in the blood at any given time. Managing **blood sugar levels** is essential for individuals with diabetes to prevent both **hypoglycemia** and hyperglycemia.

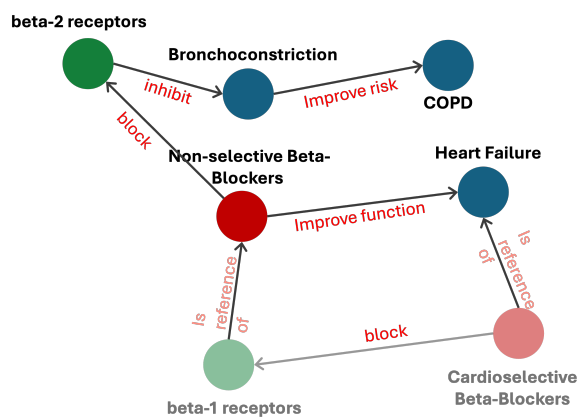
•**Fast-Acting Carbohydrates:** Carbohydrates that are rapidly absorbed into the bloodstream, quickly raising **blood glucose** levels. Examples include glucose tablets, sugary drinks, and candies. These are used to treat **hypoglycemia**.

•**Hypoglycemia:** A condition characterized by abnormally low **blood glucose** levels, which can be caused by too much insulin, insufficient food intake, or increased physical activity without proper adjustments.

Figure 15: MedGraphRAG long-form generation response on Chornic Disease Managment example-part2.



a. Abstracted graph of GraphRAG



b. Abstracted graph of MedGraphRAG

Figure 16: The comparison of abstracted graph between GraphRAG and MedGraphRAG.